Ahmed Karam Emara

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PROFESSIONAL SUMMARY

Software Engineering student with a solid foundation in Linux environments, enterprise networking, and software automation. Experienced in Agile SDLC, API integrations, and test automation, with a strong transition toward IT infrastructure, system operations, and DevOps methodologies. Proven ability to architect complex network topologies and manage headless Linux server environments.

EDUCATION

Egyptian Chinese University (ECU) *2023 – 2028 (Expected)*
B.Sc. in Software Engineering · GPA: 3.3 / 4.0 · 3rd Year Complete Cairo, Egypt

EXPERIENCE

Software Testing Trainee — 7-Month Scholarship *2025 – 2026*
Digital Egypt Pioneers Initiative (DEPI) · Ministry of Communications & IT (MCIT) Cairo, Egypt

- Delivered a comprehensive graduation project executing the complete Software Testing Life Cycle (STLC), encompassing **Manual Testing**, **API Testing (Postman)**, and **Test Automation (Java & Selenium WebDriver)**.
- Mastered Agile/Scrum methodologies and managed end-to-end defect lifecycles within **Jira**, ensuring seamless continuous integration during sprint cycles.
- Engineered robust automated test scripts and executed endpoint validation, laying the groundwork for reliable CI/CD pipeline automation.

PROJECTS

Enterprise Network Configuration (CCNA) *2025*
Networking · Cisco Packet Tracer · OSPF · VLANs Independent Project

- Architected and simulated a complete multi-site enterprise network topology connecting a Head Office (HQ) to multiple branch locations.
- Implemented Layer 3 switching, inter-VLAN routing (IT, HR, Finance, Management subnets), and configured OSPF routing protocols across all routers.
- Enforced strict access control by applying port security across all Layer 2 switches to mitigate unauthorized network access.

Autonomous System (Master-Slave Architecture) *2025–2026 | ECU Engineering*
Linux (Ubuntu) · ROS2 · Python · Hardware Integration Linux & Systems Focus

- Deployed and managed a headless Linux (Ubuntu) environment on a Raspberry Pi 5 to run ROS2 (Robot Operating System) nodes.
- Structured a 3-tier deterministic system architecture handling real-time data flow and state machine transitions.
- Root-caused and resolved hardware-to-software communication bugs via systematic hardware diagnosis and log analysis.

EasyRent — System Architecture & API Integration *2025 | ECU Coursework*
React · REST APIs · JSON · System Design Project Lead & Requirements Author

- Engineered a full-stack platform with a focus on robust API integration, connecting the frontend with Google Gemini 2.0 API for automated data processing.
- Authored formal Software Requirements Specifications (SRS), detailing system uptime requirements and module communication.

SKILLS

Technical: Linux (Ubuntu/CLI), Enterprise Networking (Routing, Switching, VLANs, OSPF), Git & GitHub, Test Automation (Selenium), API Testing (Postman), Programming (Python, Java, C++), Bash Scripting (Familiar)

Professional: Agile & Scrum Methodologies, Software Development Life Cycle (SDLC), System Troubleshooting, Log Analysis, Technical Documentation, Analytical Thinking

LANGUAGES

Arabic — Native · **English** — Advanced (C1)